Anabella Isaro

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Research interests: efficient audio representation learning, speech source separation, generative audio modeling, efficient multimodal models

EDUCATION

University of Illinois Urbana-Champaign Aug 2026 – Present
Ph.D. in Computer Science Advisor: Prof. Minje Kim

Northwestern University Jun 2023 – Dec 2024
M.S. in Artificial Intelligence GPA 3.77/4.00

Worcester Polytechnic Institute Aug 2014 – May 2018
B.S. in Computer Science GPA 3.47/4.00

RESEARCH EXPERIENCE

- **Audio Lab, University of Illinois Urbana-Champaign** Aug 2026 – Present
  Graduate Research Assistant, Advisor: Prof. Minje Kim
  - Investigating efficient audio representations for compact, low-cost speech and audio models.
  - Developing a lightweight transformer-based masking network for multi-speaker source separation, trained with a self-supervised audio large language model feature loss to preserve perceptual content.
- **Interactive Audio Lab, Northwestern University** Sep 2023 – Apr 2026
  Research Assistant, Advisor: Prof. Bryan Pardo
  - Benchmarked autoregressive, diffusion, and flow-based models for symbolic music (MIDI) generation; flow-based methods best captured long-range temporal structure.
  - Designed an evaluation framework for generated audio using large pre-trained models as references, complementing Fréchet Music Distance and KL divergence.
- **Machine Learning and Language (MLL) Lab, Northwestern University** Jan 2024 – Apr 2026
  Research Assistant, Advisor: Prof. Manling Li
  - Explored grafting multi-head latent attention into the LLM of pre-trained VLMs to reduce KV-cache memory and accelerate long-video inference without full retraining; profiled the vision encoder, prefill, and decoding stages to identify bottlenecks.
- **Emutivo Lab, Worcester Polytechnic Institute** Aug 2017 – May 2018
  Undergraduate Researcher, Advisors: Prof. Emmanuel Agu, Prof. Elke Rundensteiner
  - Helped build MOODABLE, which estimates depression from smartphone voice samples using SVMs and regression models (published in *Smart Health*).

PUBLICATIONS

- A. Dogrucu, A. Perucic, **A. Isaro**, D. Ball, E. Toto, E. A. Rundensteiner, E. Agu, R. Davis-Martin, E. Boudreaux. "Moodable: On feasibility of instantaneous depression assessment using machine learning on voice samples with retrospectively harvested smartphone and social media data." *Smart Health*, 17, 2020. doi:10.1016/j.smhl.2020.100118

FELLOWSHIPS & AWARDS

NSF CSGrad4US Graduate Research Fellowship – National Science Foundation 2026 – 2029

INDUSTRY EXPERIENCE

Tyler Technologies Mar 2019 – Jul 2026

- **Lead Software Engineer, Architecture** Nov 2025 – Jul 2026
  - Led AI enablement across engineering teams, building MCP tools and agent skills that automated repetitive development workflows.
  - Served as on-call lead developer, triaging and resolving infrastructure issues escalated by product teams.
- **Senior Software Engineer, Accounts Payable** Mar 2019 – Nov 2025
  - Led the migration of on-premises applications to AWS, rewriting legacy .NET web applications in Angular.
  - Decomposed backend services into independently deployable APIs, improving modularity and release flexibility.

SKILLS

ML & Audio: PyTorch, torchaudio, Hugging Face, Transformers, diffusion & flow models, self-supervised speech models (WavLM, HuBERT), speech LMs & neural audio codecs (Moshi/Mimi)

Languages: Python, C#, TypeScript, Java, SQL

Systems: AWS, Docker, Git, Linux, distributed training

TEACHING & TALKS

Instructor, Digital Media Academy: Intro to AI with Python; Applied Data Science & ML; OOP with Java; Mobile Development with Swift

Talk, "Optimizing Architecture and Workflow Automation in Public Sector Software," Tyler Connect 2022, Indianapolis, IN